Deliverable 1: initial domain

In this deliverable, you will perform an initial analysis of the Cairn Hiking Event Management System (CAR-HMS) and develop a high-level domain model. The system is described in the initial requirements document (CAR-Req-01).

Despite all the care taken by the requirements engineers, it is quite likely that ambiguities and gaps remain in the provided requirements. You will need to make reasonable assumptions for these cases, and document them in your report.

> **Note**
>
> All your work should be placed in the GitHub repository assigned to your group

1 Deliverable document

Create a **Markdown** document in the `docs` folder of your repository with the following outline:

1. Cover page with:
   - your team name (and optional logo), the product/project name
   - the date, course code, professor's name, team member names, **GitHub user name for each team member,** student numbers, and deliverable number.
2. Executive summary: briefly state the purpose of the document, the intended reader, and a one or two sentence summary of its contents.
3. Description of team and roles: provide a table with all team members and the tasks they were involved in for this deliverable. You can be specific by stating percentages of the work done and the number of hours spent by each team member if this information is available. If for some legitimate reason the work share is unbalanced, state the reasons in a short paragraph. The information provided here may be used to adjust team members' final grades for this deliverable.
4. Assumptions: list all assumptions made.
5. System context diagram: show an overview of the system and its interactions with other systems in its environment.
6. Sub-domains: list the sub-domains identified by your analysis of the CAR-HMS domain. For each sub-domain:
   - indicate its classification (core, supporting, generic)
   - provide the rationale for the classification (you can refer to the requirements document as well as your own interpretation)
7. Use case analysis: provide a detailed description of each of the CAR-HMS use cases. Each description must include preconditions, steps, and postconditions.
8. References.

2 Code

By the deliverable deadline, your repository on GitHub should contain the structure of the domain and the Cucumber feature files of the initial contracts (see the `contracts` branch of https://github.com/SEG3102-Fall2026/auction-system-tutorials for an example).

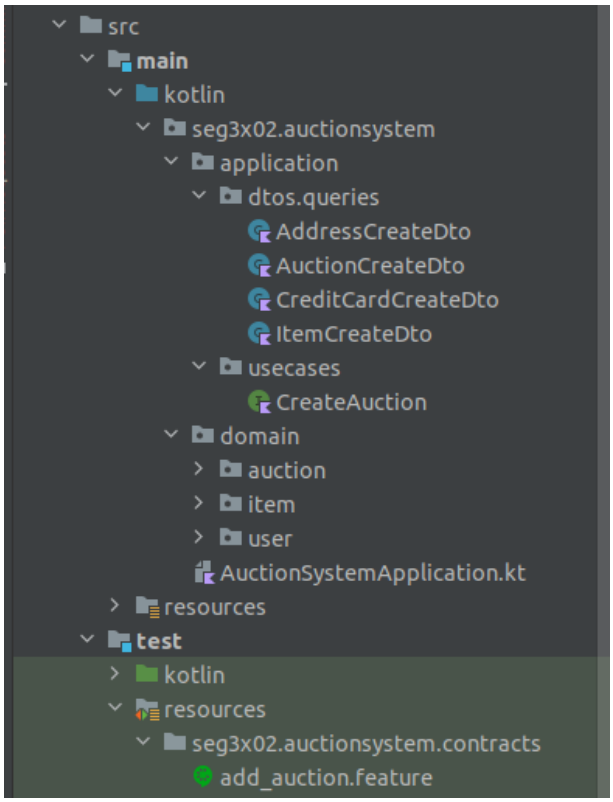


Figure 1: Layout of the contracts branch of the auction system example

- High-level structure of the domain: in a `domain` package, include a sub-package for each sub-domain listed in the sub-domains section of your document, with a brief `README.md` in each.
- Application commands: in an `application.usecases` package, include an interface for each use case except Sign In, Sign Out, and Generate Trail Usage Report. Each interface declares a function for each application command derived from the corresponding use case.
- Query data transfer objects: in an `application.dtos.queries` package, include all the DTOs used to invoke the application commands.
- Initial contracts: in a `contracts` package under the test resources, include a contract for each application command as a Cucumber feature.

The back end is written in Kotlin, as required by the club (see the technical constraints in the requirements document).

3 Bonus: requirements analysis (up to 10%)

The requirements document was written from a first meeting with the club. But it has gaps that requires you to make assumptions when coming up with the design. Your team may add a section listing gaps or inconsistencies they found, with the resolution they chose, for a bonus of up to 10% of this deliverable.

- Each item quotes the sentence of the requirements document it comes from and says what it changes in your design or in a contract.
- For each item, briefly state that if you have a chance to meet with the club again, how will you close this cap or fix this inconsistency.

- This section is separate from the assumptions section. An assumption states what you decided. This bonus item explains why the document leads to the assumption.
- At most five items are considered.

Include such a review in your deliverable document.

4 Submission

Submit your work by providing a link to the latest commit in your team's GitHub repository on Brightspace.

5 Individual contribution

Note that the work in this deliverable should be distributed equally among your team members. Your GitHub repository commit history may be used to adjust individual grade for this deliverable.

6 Marking scheme

Component	Item	Marks
Deliverable document (65 marks)	Cover page quality	5
	Executive summary	5
	Teamwork description	3
	Assumptions	5
	System context diagram	8
	Subdomain decomposition	12
	Use case analysis	25
	References	2
Code (35 marks)	Project structure	5
	Application commands	10
	Data transfer objects	5
	Initial contracts	15
Total		100
Bonus: Requirements analysis		10